

Prediction Part II

Classification, Tree, Ensemble Learning

Classification vs. Regression

From previous lectures:

Problem type	Output Space	Example Tasks
Regression	Numeric (continuous or <i>discrete</i>)	House prices, number of accidents
Classification	Categorical	Spam/non-spam, disease/no disease

A (Sometimes) Blurred Boundary

Consider the following outcomes. Are they **classification** or **regression**?

1. Hand-written digit recognition (0 to 9).
2. Housing price.
3. COVID-19 PCR test.
4. Weather forecast: probability of precipitation.

Classification Methods

Almost all methods work on numeric outcomes apply to classifications with a change in the measure/loss/objective/criteria.

More discussion in the model validation lecture.

Decision tree

From the regression lecture, you have already seen decision trees used for **numeric prediction**. Here we use the same tree idea for **classification**, then ask how to combine **many trees** into a stronger model.

Questions

- What is a decision tree? **Tree** and **decision tree**

- When and why would you use a decision tree (instead of other models)?

CART

- **CART** stands for **C**lassification **A**nd **R**egression **T**ree, introduced by Breiman, Friedman, Olshen & Stone (1984).
- CART is the standard tree-building framework students will see in many data-analysis tools: **recursively split** the feature space, then predict with the **terminal regions**.
- Classification trees usually split to make labels **purier**; regression trees usually split to reduce **within-node variation**.
- [More discussion in jupyter notebook](#)

Further reading with AI/LLM

An insightful note by **Leo Breiman** on "two cultures in the use of statistical modeling to reach conclusions from data".

In 2026, there are a few ways to read and understand a paper, for instance

1. Download and **read the paper yourself**
2. Send the paper to an **LLM** and ask questions while reading the paper

Think about the following questions in this experience

1. How do you know that you **fully understand** the paper?
2. How do you know that the **AI understands** the paper?

Ensemble Learning

Ensemble learning: methods that **aggregate multiple models** to **improve learning performance**.

"An ensemble succeeds not because every model is right, but because their mistakes do not all point the same way." — The ensemble learning mantra by GPT-5.6 Pro

"Diversity of errors is the ensemble's greatest asset: average away the noise, keep the signal." --- The ensemble learning mantra by Claude Fable 5

Tree-based Ensemble Learning

A single **decision tree** is

-  **fast and interpretable**
-  **Sensitive and prone to overfitting**

Overcome these flaws using **ensemble learning**:

Method	Main idea	What it mainly reduces
Boosting	Build trees sequentially , each fixing the last model's errors .	Bias
Bagging	Build trees in parallel on bootstrapped data ; vote/average.	Variance
Random Forest	Bagging plus feature-level randomness at each split.	Variance (even more)